Arithmetic Quantum Cat Maps:
Exact Fourth-Moment Reductions, Cayley Arithmetic, and Heisenberg-Scale Trace Criteria

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Abstract

We develop exact finite-field reductions for a fourth-moment problem arising in arithmetic quantum cat maps and isolate the arithmetic and spectral closure conditions that remain before a microscopic Bohigas–Giannoni–Schmit (BGS) statement can be attempted. For an odd prime p and a multiplicative subgroup $H \leq \mathbf{F}_p^\times$ of order K, the pair-key geometry gives the exact fourth-moment difference distribution

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0}.$$

With the Cayley transform $c(h) = (h+1)/(h-1)$ and $A = c(H \setminus \{1\})$, the mixed term is governed by $E_+(A)$, and a source-exact subgroup-curve estimate yields

$$E_+(A) < 34K^{8/3} \qquad (100 < K < p^{3/4}/3).$$

The corrected off/off elliptic layer has a coherent reducible $2p-8$ spike; after its extraction,

$$\mathrm{Var}_{t\neq 0}\, O(t) \ll pE_+(A) + K^2p + K^4p^{1/2} + K^4.$$

Thus both nontrivial fourth-moment layers reduce to the same centered Cayley energy problem.

For odd prime K, characteristic-zero cotangent rigidity gives complete additive collision rigidity and the square norm identity $\mathrm{Res}(\Phi_K, P) = K^3M^2$. We compress exceptional reductions into a canonical difference discriminant and prove a ratio-orbit fourth-power structure. A fixed-product-row decomposition reduces the pointwise excess to intersections of special affine copies of the cyclotomic trace set. After removing the projective invalid trace in the maximal real cyclotomic field, a fixed-degree subresultant coefficient ideal gives the *exact* split-prime support of row intersections of multiplicity at least three. Outside this support,

$$E_+(A) \leq 11(K-1)(K-2).$$

The remaining split arithmetic problem is to control the prime divisors of that exact real-trace ideal as K grows.

We also remove the complete-splitting assumption in a companion arithmetic family. For $p \neq 2, K$, let $f = \mathrm{ord}_K(p)$ and work over the minimal field $\mathbf{F}_{p^f}$. Every non-forced Cayley collision has a Frobenius orbit of size exactly f, so

$$f \mid \Delta_K^{\mathrm{ext}}(p).$$

The collision-resultant budget and canonical discriminant extend to all such rational primes. Consequently, for unrestricted prime pairs $K \asymp X^\theta$, $p \asymp X$, asymptotically almost every pair has the exact Cayley baseline in its minimal extension for every fixed $\theta < 1/3$. This does not strengthen the split slice itself, where $f = 1$.

Finally, we close the deterministic Fourier part of the microscopic local-statistics problem. Möbius inversion on the partition lattice expresses distinct-index spectral exponential

sums exactly through joint products of traces, and Poisson summation expresses factorial microscopic k-point statistics by these combinations. Band-limited microscopic tests therefore require traces at powers $|n| = \Theta(N)$, the Heisenberg scale. The missing spectral theorem is a weighted joint-trace comparison for the correct desymmetrized block (plus a high-frequency tail estimate beyond band-limited tests). No microscopic BGS/GOE/COE universality is claimed.

1 Scope and evidence discipline

Throughout the split finite-field part, p is an odd prime and $H \leq \mathbf{F}_p^\times$ is a multiplicative subgroup of order K. Division is in $\mathbf{F}_p$. Universal claims are labelled exact only when the load-bearing derivation is present; finite computations are regression or certificates, never substitutes for proof. When a theorem from the literature is load-bearing, we identify the source and the hypotheses used.

The distinction between moment information and microscopic spectral statistics is essential. Arithmetic cat-map literature contains strong eigenfunction, matrix-element and quantum-ergodicity results [1, 2, 3, 4, 5], but these do not imply a local eigenphase law. Conversely, rigorous random-matrix universality is known for special model classes such as mixing quantum graphs under their own structural and large-graph assumptions [6, 7]; no transfer to the arithmetic family studied here is assumed.

2 Exact pair-key geometry and fourth-moment coupling

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every such (u, v) gives a unique admissible off-diagonal pair. The diagonal $x = y$ collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. The equations $x = uy$ and $x - 1 = v(y - 1)$ give $(u - v)y = 1 - v$, and then (2). The diagonal statement is immediate. □

Corollary 2.2. *Under the two subgroup constraints $u, v \in H$, the number of admissible off-diagonal ordered pairs is exactly $(K - 1)(K - 2)$.*

Give each collapsed key $r = (u_r, v_r, d_r)$ multiplicity n_r, equal to one off the diagonal and $p - 2$ at $(1, 1, 0)$. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \mathbf{1}_H(v_r v_s) \mathbf{1}_{d_r + d_s = t}. \tag{3}$$

Let $O(t)$ be the off/off contribution and

$$C_H(t) = \#\left\{(u, v) \in (H \setminus \{1\})^2 : u \neq v, \ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

Then

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2 \mathbf{1}_{t=0}. \tag{4}$$

The coefficient 2 is essential: diagonal/off and off/diagonal are distinct ordered classes.

Proposition 2.3 (Exact total coupled count)**.**

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K.$$

Let $S = H^\perp$ be the multiplicative-character annihilator and $m = |S| = (p-1)/K$. In the maintained twisted-Jacobi convention,

$$\sum_{\chi,\psi\in S} |J(\chi,\psi;q)|^4 = m^2\Sigma_4(q), \qquad \Sigma_4(q) = \sum_{t\in\mathbf{F}_p} B(t)e_p(qt). \tag{5}$$

Writing $V = \sum_t B(t)$ and $Z = B(0)$, finite Fourier orthogonality gives

$$\mathbb{E}_{q\neq 0}\Sigma_4(q) = \frac{pZ - V}{p-1}, \qquad \mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = p\,\mathrm{Var}_{t\neq 0}\,B(t). \tag{6}$$

No probabilistic independence assumption enters this step.

3 Cayley transform and the pointwise energy target

Put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H\setminus\{1\}).$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u)-c(v)}, \qquad c(h^{-1}) = -c(h). \tag{7}$$

Thus $A = -A$, $0\notin A$, and the nonzero multiplicities of C_H are the nonzero additive-difference multiplicities of A, up to the permutation $x\mapsto 2/x$. Consequently

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{8}$$

With $n = |A| = K-1$, identity and central inversion force the nonzero baseline

$$n(2n-3) = (K-1)(2K-5). \tag{9}$$

For every $A\subset\mathbf{F}_p$, with the unnormalized Fourier transform $\widehat{1_A}(q) = \sum_{a\in A} e_p(qa)$, Parseval gives

$$E_+(A) = \frac{|A|^4}{p} + \frac{1}{p}\sum_{q\neq 0} |\widehat{1_A}(q)|^4. \tag{10}$$

Hence the natural pointwise target is

$$E_+(A) = \frac{(K-1)^4}{p} + O(K^{2+o(1)}), \tag{11}$$

which is equivalent to

$$\sum_{q\neq 0} |\widehat{1_A}(q)|^4 \ll pK^{2+o(1)}.$$

This is the principal open split-prime arithmetic target.

There are two useful exact reformulations. First, with $y = (h-1)^{-1}$,

$$h = \frac{y+1}{y}, \qquad c(h) = 1+2y,$$

so (11) is an additive-energy problem for the reciprocal shifted subgroup $(H-1)^{-1}$. Second, solving $c(h)-c(k)=t$ gives

$$k=\frac{(2-t)h+t}{2+t-th}, \tag{12}$$

which is the Cayley conjugate of a translation; the problem may therefore be viewed as an L^2 intersection problem between a multiplicative torus and a one-parameter unipotent family. Known shifted-subgroup multiplicative-energy and Möbius-incidence estimates [13, 14] do not by themselves imply (11).

4 A uniform pointwise bound

For $t\in\mathbf{F}_p$, write $r(t)=\#\{(a,b)\in A^2 : a+b=t\}$. If $a=c(x)$ and $b=c(y)$, clearing Cayley denominators gives

$$P_t(x,y)=(2-t)xy+t(x+y)-(2+t)=0. \tag{13}$$

The bilinear determinant is

$$(2-t)(-(2+t))-t^2=-4,$$

so every fiber is absolutely irreducible in odd characteristic.

Theorem 4.1 (Direct Cayley energy bound)**.** *If $100<K<\frac{1}{3}p^{3/4}$, then*

$$E_+(A)<34K^{8/3}. \tag{14}$$

Proof. For $t\neq 0,-2$, the bidegree-$(1,1)$ polynomial (13) satisfies $P_t(0,0)\neq 0$ and $\deg P_t(x,0)=1$. Theorem 2 of the source version of Vyugin–Makarychev [11], under the displayed range, gives $r(t)\le 32K^{2/3}$. The exceptional fibers satisfy $r(0)=K-1$ and trivially $r(-2)\le K-1$. Since $\sum_t r(t)=(K-1)^2$,

$$E_+(A)=\sum_t r(t)^2\le 2(K-1)^2+32K^{2/3}\sum_{t\neq 0,-2} r(t)<34K^{8/3}.$$

□

The later journal publication is [12]. The source hypothesis $\deg P(x,0)\ge 1$ was checked directly. Combining (8) with the coefficient $2(p-2)$ in (4) gives

$$\mathrm{Var}_{t\neq 0}(2(p-2)C_H(t))<136pK^{8/3}. \tag{15}$$

5 Prime-order characteristic-zero rigidity and square resultants

Let K now be an odd prime and $\zeta=e^{2\pi i/K}$. Then

$$c(\zeta^a)=-i\cot\frac{\pi a}{K}. \tag{16}$$

Chowla proved that the $(K-1)/2$ cotangents with $1\le a\le (K-1)/2$ are rationally independent [8]; Wang states the prime case explicitly and extends it [9].

Lemma 5.1 (Cross-polytope differences)**.** *For vertices x,y,u,v of a rational cross-polytope, with $x\neq y$ and $u\neq v$, the equality $x-y=u-v$ implies either $(u,v)=(x,y)$ or $(u,v)=(-y,-x)$.*

Proof. Write $x+v=u+y$. If the common sum vanishes, the second alternative holds. Otherwise a sum of two signed basis vectors determines its unordered summands, and the other pairing would force $x=y$. □

Theorem 5.2 (Prime-order additive rigidity)**.** *Every characteristic-zero equal-difference collision among* $\{c(\zeta^a) : a \in \mathbf{F}_K^*\}$ *is identity or central inversion-swap.*

For an admissible collision quadruple, let $P(X)$ be the integral polynomial obtained by clearing the four Cayley denominators. Direct algebra gives

$$P(X) = (X-1)^3 Q(X), \qquad Q(X^{-1}) = X^{3-T} Q(X)$$

for the appropriate exponent sum T, while $\|P\|_1 \le 12$.

Theorem 5.3 (Square-resultant theorem)**.** *For every odd prime* K *and every admissible collision quadruple,*

$$\mathrm{Res}(\Phi_K, P) = K^3 M^2, \qquad M \in \mathbf{Z}. \tag{17}$$

For a non-forced characteristic-zero collision, $M \neq 0$.

Proof. Put $L = \mathbf{Q}(\zeta)$ and let L^+ be its maximal real subfield. Choose m with $2m \equiv 3 - T \pmod K$. The reciprocal identity makes $\beta = \zeta^m Q(\zeta)$ real, hence an algebraic integer of L^+. Therefore

$$\mathrm{N}_{L/\mathbf{Q}}(Q(\zeta)) = \mathrm{N}_{L^+/\mathbf{Q}}(\beta)^2 = M^2.$$

Since $\mathrm{N}_{L/\mathbf{Q}}(\zeta - 1) = K$, taking norms of $P(\zeta) = (\zeta-1)^3 Q(\zeta)$ proves (17). Non-vanishing follows from Theorem 5.2. □

A non-forced collision modulo a split prime $p \equiv 1 \pmod K$ implies $p \mid M$ and hence

$$K^3 p^2 \le 12^{K-1}. \tag{18}$$

This improves the elementary cutoff but remains exponential in K.

6 Fibonacci slice and a large finite certificate

The relation

$$c(h^2) - c(h) = c(h) - c(h^{-1})$$

is equivalent, for odd-order $h \neq 1$, to $h^2 + 3h + 1 = 0$.

Theorem 6.1 (Fibonacci resultant)**.** *For every odd prime* K*,*

$$\mathrm{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2.$$

The corresponding cleared Cayley collision has full norm $K^3 F_K^2$.

Proof. If α, β are the roots of $X^2 + 3X + 1$, then $\alpha\beta = 1$ and $\alpha + \beta = -3$. Evaluating $\Phi_K(X) = (X^K - 1)/(X - 1)$ at the two roots reduces the resultant to $(2 - (\alpha^K + \beta^K))/5$. Writing the roots as $-\varphi^2, -\varphi^{-2}$ and using $L_{2K} = 5F_K^2 - 2$ for odd K yields the identity. □

Thus total collision norms may be exponentially large, so "remove every exceptional prime" is the wrong closure condition. A stronger structural fact is that the rational Fibonacci/Lucas model

$$S_N = \{\pm L_n / F_n : 1 \le n \le N\}$$

has exact nonzero difference energy

$$E^*(S_N) = 2N(4N-3) + 32 \qquad (N \ge 3). \tag{19}$$

The unbounded divisibility separation uses the classical Fibonacci primitive-divisor theorem in the source-exact form of Yabuta [10]; the bounded residue is checked independently in Lean.

At the exact finite-field design

$$K = 131, \qquad p = F_{131} = 1066340417491710595814572169,$$

a complete Lucas primality certificate and independent Python/C++ arbitrary-precision computations give nonzero Cayley energy 33442, forced baseline 33410, and excess 32. This is a finite certificate, not an asymptotic theorem.

7 Canonical difference discriminant

Let

$$Q_K(X) = \frac{(X+1)^K - (X-1)^K}{2}.$$

Its roots are exactly the characteristic-zero Cayley set. Put

$$R_K(T) = \mathrm{Res}_X(Q_K(X), Q_K(X-T)),$$

remove the diagonal factor T^{K-1}, and let $S_K(T)$ be the primitive squarefree part. By Theorem 5.2, each nonzero characteristic-zero difference root has weight one or two, and

$$\deg S_K = \frac{(K-1)^2}{2}.$$

Theorem 7.1 (Single-discriminant classification)**.** *For every odd prime K and every prime $p \equiv 1 \pmod K$,*

$$\Delta_K(p) > 0 \iff p \mid \mathrm{Disc}(S_K), \qquad \Delta_K(p) \le 4 v_p(\mathrm{Disc}(S_K)), \tag{20}$$

where

$$\Delta_K(p) = \sum_{d \neq 0} r_{A-A}(d)^2 - (K-1)(2K-5).$$

Proof. The leading coefficient of S_K is a p-adic unit and all its roots lie in $\mathbf{Q}_p$. Hence

$$v_p(\mathrm{Disc}\, S_K) = 2 \sum_{\alpha<\beta} v_p(\alpha - \beta).$$

Each colliding reduction pair contributes at least two to this valuation. The collision weights are at most two, giving the factor 4. □

8 Ratio-orbit factorization and a fourth-power discriminant

For $r \in \mathbf{F}_K^* \setminus \{1\}$ and $a \in \mathbf{F}_K^*$ define

$$d_{r,a} = c(\zeta^a) - c(\zeta^{ra}).$$

The involution $(a, ra) \mapsto (-ra, -a)$ identifies r with r^{-1}, giving $(K-1)/2$ ratio classes. For one class set

$$D_r(X) = (X-1)(X^r - 1), \qquad N_r(X) = 2(X^r - X),$$

$$F_r(T) = \frac{1}{K} \mathrm{Res}_X\big(\Phi_K(X), T D_r(X) - N_r(X)\big).$$

Then $F_r = F_{r^{-1}}$, distinct ratio classes give coprime factors, and

$$S_K(T) = \prod_{[r]} F_r(T). \tag{21}$$

After the even rescaling $F_r(2U) = 2^{K-1}H_r(U^2)$ one obtains $H_r(Y) = 1 + KP_r(Y)$ with P_r monic. The next coefficient is

$$\frac{(K-1)(K-2)}{12} - Ks(r,K),$$

where $s(r,K)$ is the Dedekind sum. The prime-modulus equality criterion [15] distinguishes the ratio classes and yields an exact square in every cross-resultant.

Theorem 8.1 (Fourth-power discriminant structure)**.** *Let $n = K-1$. For each ratio class there is an integer $I_r > 0$ such that*

$$|\operatorname{Disc}(F_r)| = 2^{n^2}K^{K-2}I_r^4.$$

For distinct classes there is an integer $M_{r,s} \neq 0$ such that

$$|\operatorname{Res}(F_r, F_s)| = 2^{n^2}K^{K+1}M_{r,s}^2.$$

Consequently, for a positive integer J_K,

$$|\operatorname{Disc}(S_K)| = 2^{(K-1)^4/4}K^{(K-1)(K^2-7)/4}J_K^4. \tag{22}$$

Exact ratio-orbit computations through $K = 47$ reject the earlier small-K heuristic $O(K^3 \log K)$ for the residual logarithmic mass. Cross-resultants retain genuine K^4-scale archimedean mass. Thus (22) is structural, not a hidden proof of (11).

9 Multiplicative Cayley rigidity

The same Cayley set has an independent multiplicative rigidity. If $h_1 = z/v$ and $h_2 = zv$, write $T_x = x + x^{-1}$. Then

$$c(z/v)c(zv) = \frac{T_z + T_v}{T_z - T_v}.$$

Equality of two such products reduces to equality of four-point exponent multisets; prime cyclotomic linear independence leaves only signed permutations.

Theorem 9.1 (Multiplicative rigidity)**.** *For $a, b, c, d \in \mathbf{F}_K^*$,*

$$c(\zeta^a)c(\zeta^b) = c(\zeta^c)c(\zeta^d)$$

if and only if (c,d) is one of

$$(a,b),\ (b,a),\ (-a,-b),\ (-b,-a).$$

Consequently

$$E_\times(A_K) = 4(K-1)(K-2).$$

There is a canonical squarefree product polynomial $S_K^\times$ with the same reduction-cluster type exceptional-prime classification. No implication from multiplicative to additive energy is assumed.

10 Fixed-product rows and exact split order-three support

The additive-energy excess has a second exact organization that is sharper for pointwise control. For a valid pair put $u = xy$ and $s = x + y$. Direct calculation gives

$$c(x) + c(y) = \frac{2(u-1)}{u-s+1}. \tag{23}$$

At fixed $u \neq 1$ and fixed nonzero Cayley parameter, the sum s is fixed and there is at most one unordered pair $\{x, y\} \subset H \setminus \{1\}$. Let $I_K(p)$ be the maximum number of common parameters of two distinct product rows. The exact cross-row decomposition gives

$$\Delta_K(p) \leq 4I_K(p)(K-1)(K-2). \tag{24}$$

Normalize a distinct row pair to $w = \zeta$, $q = \zeta^r$, put $\tau = \zeta + \zeta^{-1}$, and let U_j denote the Chebyshev polynomial of the second kind. The constrained affine trace map is

$$Y = \alpha_r X + \beta_r, \qquad \alpha_r = U_{r-1}(\tau/2), \quad \beta_r = -2U_{r-2}(\tau/2). \tag{25}$$

Let $C_K(T)$ be the minimal polynomial of τ in the maximal real cyclotomic field and put

$$H_K(T) = (T-2)C_K(T).$$

The unique characteristic-zero common trace is the invalid row trace $T = \tau$. Remove it before reduction:

$$P_{K,r}(T) = \frac{H_K(T)}{T-\tau}, \qquad Q_{K,r}(T) = \frac{H_K(\alpha_r T + \beta_r)}{T-\tau}. \tag{26}$$

Both have fixed degree $m = (K-1)/2$ over $\mathbf{Z}[\tau]$. For every split $p \neq K$, reduction preserves the degree and squarefreeness, and the gcd degree of (26) equals the number of common Cayley parameters of the corresponding row pair.

Let $J^{\mathrm{tr}}_{K,r}$ be the ideal generated by all coefficients of the first three fixed-degree subresultants of $P_{K,r}, Q_{K,r}$ and define

$$J^{\mathrm{tr}}_{K,r} \cap \mathbf{Z} = \Lambda^{(3)}_{K,r}\mathbf{Z}.$$

Theorem 10.1 (Exact real-trace order-three support)**.** *For every odd prime K and split prime $p \equiv 1 \pmod K$,*

$$p \mid \Lambda^{(3)}_{K,r}$$

if and only if, for some row pair in the ratio class $[r]$, the finite-field row intersection contains at least three common Cayley parameters.

Proof sketch. The cofactors in (26) keep fixed degree after split reduction and their roots are exactly the valid row trace classes. Hence the geometric intersection count equals their gcd degree. The fixed-degree subresultant criterion [16] says gcd degree at least three is equivalent to vanishing of the first three subresultants. Requiring all their coefficients to vanish in one prime ideal above p is exactly the support of the coefficient ideal; contracting to $\mathbf{Z}$ gives the rational-prime statement. □

After deleting the irrelevant factors $2, K$ and multiplying over ratio classes, write the resulting exact support integer as ${\Lambda'_K}^{(3)}$. Then

$$p \nmid {\Lambda'_K}^{(3)} \quad \Longrightarrow \quad I_K(p) \leq 2, \quad E_+(A) \leq 11(K-1)(K-2). \tag{27}$$

The growing-parameter target

$$P^+_{\mathrm{split}}({\Lambda'_K}^{(3)}) \leq K^{4+o(1)} \tag{28}$$

would close the pointwise Cayley-energy problem throughout $K = o(p^{1/4})$.

Remark 10.2 (Scope correction)**.** *Earlier dehomogenized projective-row integers* ${D'_{K,r}}^{(3)}$ *and* ${G'_{K,r}}^{(3)}$ *remain valid one-way obstructions*

$$I \geq 3 \Longrightarrow p \mid D' \Longrightarrow p \mid G'.$$

Their converses are false in general because specialization can lower the leading degree at the projective point $\mu = \infty$*. The real-trace cofactors were introduced to remove exactly this degree-drop ambiguity.*

11 A global split-prime collision budget

Simultaneous multiplication of all four exponent indices by $a \in \mathbf{F}_K^\times$ acts freely on non-forced collision quadruples. The number of normalized scaling orbits is

$$N_K = K^3 - 5K^2 + 6K + 1. \tag{29}$$

For one normalized orbit q, let P_q be its collision polynomial and $R_q = \mathrm{Res}(\Phi_K, P_q)$.

Theorem 11.1 (Weighted split-prime budget)**.** *For every odd prime* K*,*

$$\sum_{p \equiv 1\,(K)} \Delta_K(p) \log p \leq N_K\big((K-1)\log 12 - 3\log K\big). \tag{30}$$

Proof. For a split prime, the number of scaled members of one orbit that collide is bounded by $v_p(R_q)$. Since $R_q = K^3 M_q^2$,

$$\sum_p \Delta_K(p) \log p \leq \sum_q 2 \log |M_q| = \sum_q \log \frac{|R_q|}{K^3},$$

and $|R_q| \leq 12^{K-1}$. □

12 Frobenius amplification over minimal finite extensions

The preceding budget does not actually require complete splitting once the Cayley set is placed in the minimal residue field. Let $p \neq 2, K$ be any rational prime and put

$$f = \mathrm{ord}_K(p), \qquad q = p^f.$$

Then $\mathbf{F}_q^\times$ contains the unique subgroup H of order K. Define $A_{K,p} = c(H \setminus \{1\})$ and let $\Delta_K^{\mathrm{ext}}(p)$ denote the non-forced nonzero equal-difference excess over $\mathbf{F}_q$.

Theorem 12.1 (Frobenius amplification)**.** *Every non-forced collision orbit under the* p*-power Frobenius has size exactly* f*. In particular,*

$$f \mid \Delta_K^{\mathrm{ext}}(p). \tag{31}$$

Proof. Index the K-th roots by h^i, $i \in \mathbf{F}_K^*$. Frobenius sends a collision tuple (i, j, r, s) to (pi, pj, pr, ps) modulo K. If its dth power, $0 < d < f$, fixed the tuple, then from any nonzero coordinate $p^d i \equiv i \pmod K$ we would obtain $p^d \equiv 1 \pmod K$, contradicting the definition of f. □

Choose one prime ideal $\mathfrak{P}$ over p in $\mathbf{Z}[\zeta]$. The normalized collision resultant is a rational integer and $p \neq K$ is unramified. Every scaled collision root modulo $\mathfrak{P}$ contributes at least one to the $\mathfrak{P}$-valuation, which equals the ordinary p-valuation on rational integers. Thus the proof of Theorem 11.1 extends verbatim:

Theorem 12.2 (All-prime minimal-extension budget)**.** *For every odd prime K,*

$$\sum_{p\neq 2,K} \Delta_K^{\mathrm{ext}}(p)\log p \le N_K\big((K-1)\log 12 - 3\log K\big). \tag{32}$$

Moreover, for every $p \neq 2, K$,

$$\Delta_K^{\mathrm{ext}}(p) > 0 \iff p \mid \mathrm{Disc}(S_K), \qquad \Delta_K^{\mathrm{ext}}(p) \le 4v_p(\mathrm{Disc}(S_K)). \tag{33}$$

The Frobenius factor produces a stronger exact-baseline density theorem for a different arithmetic family. Let $Q = X^\theta$ and

$$\mathcal{D}_\theta^{\mathrm{ext}}(X) = \{(K,p) : K,p \text{ prime},\ Q/2 < K \le Q,\ X/2 < p \le X\},$$

with no congruence condition on p.

Theorem 12.3 (Density-one minimal-extension baseline)**.** *For every fixed $0 < \theta < 1/3$,*

$$\frac{\#\{(K,p) \in \mathcal{D}_\theta^{\mathrm{ext}}(X) : \Delta_K^{\mathrm{ext}}(p) > 0\}}{|\mathcal{D}_\theta^{\mathrm{ext}}(X)|} \longrightarrow 0.$$

Hence asymptotically almost every pair satisfies

$$\Delta_K^{\mathrm{ext}}(p) = 0, \qquad E_+(A_{K,p}) = 3(K-1)(K-2)$$

exactly.

Proof sketch. Choose β with $\max(0, 4-1/\theta) < \beta < 1$. Residue classes modulo K of multiplicative order at most K^β number at most $K^\beta\tau(K-1) = K^{\beta+o(1)}$; Brun–Titchmarsh and PNT show that such prime pairs have density $O(Q^{\beta-1+o(1)})$. On the complement, Theorem 12.1 makes every bad prime cost more than K^β units of excess. Applying (32) gives density $O(Q^{4-\beta}/X)$. Both terms vanish by the choice of β. □

Theorem 12.3 does not improve the original split family: $p \equiv 1 \pmod K$ has $f = 1$, so the Frobenius amplification disappears there.

13 Full Jacobi trace and split two-parameter averages

For $\chi, \psi \in S = H^\perp$ define

$$J_S(q)_{\chi,\psi} = \sum_{x\in\mathbf{F}_p\setminus\{0,1\}} \chi^{-1}(x)\psi(x-1)e_p(qx), \qquad M_S(q) = \frac{e_p(-q/2)}{p-1} J_S(q).$$

Theorem 13.1 (Cayley–Jacobi trace identity)**.** *For every $r \in \mathbf{F}_p$,*

$$\widehat{1_A}(r) = K \operatorname{Tr} M_S(2r). \tag{34}$$

Consequently

$$\sum_q |\operatorname{Tr} M_S(q)|^2 = \frac{p(K-1)}{K^2}, \tag{35}$$

$$\sum_q |\operatorname{Tr} M_S(q)|^4 = \frac{pE_+(A)}{K^4}. \tag{36}$$

Proof. Put $y = (h-1)^{-1}$ and use character orthogonality on H. After the change $x = y+1$, the resulting sums are the diagonal entries of $J_S(2r)$ up to the normalizing phase. Summing over the annihilator gives (34); Parseval gives (35)–(36). □

For $q \neq 0$, the exact moments are

$$\mu_2(K,p) = \frac{(K-1)(p-K+1)}{K^2(p-1)}, \tag{37}$$

$$\mu_4(K,p) = \frac{pE_+(A) - (K-1)^4}{K^4(p-1)}. \tag{38}$$

Fix $0 < \theta < 1/2$, put $Q = X^\theta$, and let

$$\mathcal{D}_\theta(X) = \{(K,p) : K,p \text{ prime},\ Q/2 < K \le Q,\ X/2 < p \le X,\ p \equiv 1 \pmod{K}\}.$$

Bombieri–Vinogradov and PNT [17, 18] give

$$|\mathcal{D}_\theta(X)| \sim \frac{\log 2}{2\theta} \frac{X}{(\log X)^2} \tag{39}$$

for fixed $\theta < 1/2$.

Theorem 13.2 (Double-averaged quadratic excess)**.** *For every fixed $0 < \theta < 1/3$,*

$$\frac{1}{|\mathcal{D}_\theta(X)|} \sum_{(K,p)\in\mathcal{D}_\theta(X)} \frac{\Delta_K(p)}{K^2} \ll_\theta X^{3\theta-1}. \tag{40}$$

Via (37)–(38), the standardized nonzero-frequency fourth/second moment ratio has double-average absolute deviation from 3 tending to zero for $\theta < 1/3$. This is a fourth-moment statement, not a central limit theorem.

14 Density-one exact split baseline below $\theta = 1/5$

The excess $\Delta_K(p)$ is a nonnegative integer.

Theorem 14.1 (Density-one exact split Cayley baseline)**.** *For every fixed $0 < \theta < 1/2$, let*

$$\mathcal{B}_\theta(X) = \{(K,p) \in \mathcal{D}_\theta(X) : \Delta_K(p) > 0\}.$$

Then

$$|\mathcal{B}_\theta(X)| \ll_\theta \frac{X^{5\theta}}{(\log X)^2}, \qquad \frac{|\mathcal{B}_\theta(X)|}{|\mathcal{D}_\theta(X)|} \ll_\theta X^{5\theta-1}. \tag{41}$$

In particular, for fixed $0 < \theta < 1/5$, asymptotically almost every split pair satisfies

$$\Delta_K(p) = 0, \qquad E_+(A) = 3(K-1)(K-2) \tag{42}$$

exactly.

On this density-one subfamily the exact trace kurtosis is

$$\kappa(K,p) = \frac{(p-1)\,[3p(K-2) - (K-1)^3]}{(K-1)(p-K+1)^2} = 3 + O\!\left(\frac{1}{K} + \frac{K^2}{p}\right).$$

Again, this is not a distributional or local-eigenvalue theorem.

15 Corrected off/off elliptic correlation

For an off-diagonal key (u, v) set

$$d(u, v) = \frac{(u-1)(v-1)}{v-u}.$$

For $\lambda = (h_1, h_2) \in H^2$, let $O_\lambda(t)$ count valid off-diagonal (u, v) such that $(h_1/u, h_2/v)$ is also off-diagonal and $d(u, v) + d(h_1/u, h_2/v) = t$. Then $O(t) = \sum_\lambda O_\lambda(t)$.

Assume first $h_1, h_2 \neq 1$, $h_1 \neq h_2$, $h_1 h_2 \neq 1$. Put

$$\alpha = c(h_1), \quad \beta = c(h_2), \quad s = \alpha + \beta, \quad d = \alpha - \beta.$$

With $a = c(u)$, $b = c(v)$ and $T_\alpha(a) = (\alpha a - 1)/(a - \alpha)$, the fiber equation is

$$\frac{2}{a-b} + \frac{2}{T_\alpha(a) - T_\beta(b)} = t. \tag{43}$$

Clearing denominators and taking the branch-quartic discriminant gives, up to an everywhere nonzero scalar,

$$\mathrm{Disc}_a(q) = d^2(dt-2)^4(s-d-2)^2(s-d+2)^2(s+d-2)^2(s+d+2)^2 R_{s,d}(t), \tag{44}$$

$$R_{s,d}(t) = d^2t^4 + 2d(s^2-d^2-4)t^3 + 4(5d^2-s^2+4)t^2 - 64dt + 64. \tag{45}$$

The residual discriminant is

$$\mathrm{Disc}_t\, R_{s,d} = 2^{16} d^2 s^2 (s-d-2)^2(s-d+2)^2(s+d-2)^2(s+d+2)^2. \tag{46}$$

The generic family has four simple residual bad fibers and the distinguished fiber $t_\lambda = 2/d$. At this point the cleared cubic factors into two distinct affine lines. After valid-domain exclusions their union contains exactly

$$O_\lambda(t_\lambda) = 2p - 8 \tag{47}$$

valid points.

Let $a_\lambda(t)$ be the unnormalized Frobenius trace of the Jacobian of the generic genus-one pencil. Boundary corrections have bounded degree, so

$$O_\lambda(t) = p + p\mathbf{1}_{t=t_\lambda} - a_\lambda(t) + e_\lambda(t), \qquad |e_\lambda(t)| \leq C. \tag{48}$$

The pole of the j-invariant proves non-isotriviality. Igusa open image, in the form of [20], gives geometric irreducibility. Equality of singular supports recovers the parameters up to the exact involution $(h_1, h_2) \leftrightarrow (h_2^{-1}, h_1^{-1})$. For other generic parameter pairs, the cohomological argument of Fouvry–Kowalski–Michel [19] gives

$$\sum_{t \in \mathbf{F}_p} a_\lambda(t)\overline{a_\mu(t)} = O(p^{3/2}). \tag{49}$$

Define $D(t) = \#\{\lambda \text{ generic} : t_\lambda = t\}$. Since $d \mapsto 2/d$ permutes $\mathbf{F}_p^*$ and $d = \alpha - \beta$,

$$\sum_{t \neq 0} D(t)^2 \leq E_+(A). \tag{50}$$

Theorem 15.1 (Corrected off/off variance)**.** *Let $p > 3$ and let $H \leq \mathbf{F}_p^\times$ have odd order K. Then*

$$\mathrm{Var}_{t \neq 0}\, O(t) \ll pE_+(A) + K^2 p + K^4 p^{1/2} + K^4, \tag{51}$$

with an absolute implied constant.

Combining Theorems 4.1 and 15.1 gives

$$\mathrm{Var}_{t \neq 0}\, O(t) \ll pK^{8/3} + K^4 p^{1/2} + K^2 p + K^4. \tag{52}$$

In particular, if $K = o(p^{1/4})$, then $\mathrm{Var}\, O \ll pK^{8/3}$.

16 Exact trace-to-local-statistics bridge

The preceding arithmetic trace identity is still not microscopic spectral statistics. We now isolate the deterministic Fourier conversion needed at the spectral level.

Let U be an $N \times N$ unitary matrix with eigenvalues $e^{2\pi i\theta_1}, \dots, e^{2\pi i\theta_N}$. For a zero-sum integer tuple $m_1 + \cdots + m_k = 0$, define

$$D_U(m_1, \dots, m_k) = \sum_{j_1,\dots,j_k \ mdistinct} e^{2\pi i \sum_{a=1}^{k} m_a \theta_{j_a}}.$$

For a set partition π of $\{1, \dots, k\}$, put

$$\mu(\pi) = \prod_{B\in\pi} (-1)^{|B|-1}(|B|-1)!, \qquad T_U(n) = \operatorname{Tr}(U^n), \quad T_U(0) = N.$$

Theorem 16.1 (Factorial trace identity)**.** *For every integer zero-sum k-tuple,*

$$D_U(m_1, \dots, m_k) = \sum_{\pi} \mu(\pi) \prod_{B\in\pi} T_U\left(\sum_{a\in B} m_a\right). \tag{53}$$

Proof. For a partition π, impose equality of eigenvalue indices inside each block. The corresponding unrestricted sum is the product on the right side of (53). Requiring all indices to be distinct selects the minimal partition; Möbius inversion on the partition lattice has exactly the coefficient $\mu(\pi)$ above. □

Let $\phi \in \mathcal{S}(\mathbf{R}^{k-1})$ with Fourier convention $\widehat{\phi}(\xi) = \int \phi(x) e^{-2\pi i x\cdot\xi}\, dx$, and periodize at the mean-spacing scale:

$$\Phi_N(t) = \sum_{n\in\mathbf{Z}^{k-1}} \phi(N(t+n)).$$

Define

$$C_{k,N}(U;\phi) = \frac{1}{N} \sum_{j_1,\dots,j_k \ mdistinct} \Phi_N(\theta_{j_1} - \theta_{j_k}, \dots, \theta_{j_{k-1}} - \theta_{j_k}).$$

Theorem 16.2 (Trace-to-local-statistics identity)**.** *With $m_k = -(m_1 + \cdots + m_{k-1})$,*

$$C_{k,N}(U;\phi) = N^{-k} \sum_{m\in\mathbf{Z}^{k-1}} \widehat{\phi}(m/N) D_U(m_1, \dots, m_k). \tag{54}$$

Proof. Poisson summation gives $\Phi_N(t) = N^{-(k-1)} \sum_m \widehat{\phi}(m/N) e^{2\pi i m\cdot t}$. Insert this in the definition of $C_{k,N}$ and interchange the absolutely convergent Schwartz sum with the finite eigenvalue sum. □

For $k = 2$,

$$C_{2,N}(U;\phi) = N^{-2} \sum_{m\in\mathbf{Z}} \widehat{\phi}(m/N)(|\operatorname{Tr} U^m|^2 - N). \tag{55}$$

If $\widehat{\phi}$ has compact support, then (54) involves powers of order N. Hence microscopic band-limited statistics require joint traces at the Heisenberg scale.

Corollary 16.3 (Weighted Heisenberg trace criterion)**.** *Let $\mathbb{E}_N$ be the averaging functional for the quantum system and $\mathbb{E}_N^{\mathrm{RMT}}$ the matching random-matrix average. If for a fixed band-limited ϕ*

$$N^{-k} \sum_m |\widehat{\phi}(m/N)| \left|\mathbb{E}_N D_U(m) - \mathbb{E}_N^{\mathrm{RMT}} D_V(m)\right| \longrightarrow 0,$$

then the corresponding averaged microscopic factorial statistic converges to its RMT value.

For arbitrary Schwartz tests one additionally needs a uniform high-frequency tail estimate. The deterministic Fourier bridge is therefore closed; the hard spectral task is the Heisenberg-scale joint-trace comparison for the correct desymmetrized block.

17 The remaining BGS frontier

There are now two precise, logically separate closure chains.

The arithmetic fourth-moment chain is

$$\boxed{\text{exact real-trace split support}} \longrightarrow \boxed{\text{pointwise centered Cayley energy}} \longrightarrow \boxed{\text{full fourth-moment control}}.$$

The first unresolved step may be attacked through the exact support integer ${\Lambda'_K}^{(3)}$ and its Chebyshev/cyclotomic-unit structure. The best unconditional pointwise exponent remains 8/3.

The microscopic spectral chain is

$$\boxed{\text{Heisenberg-scale desymmetrized joint traces}} \longrightarrow \boxed{\text{band-limited microscopic RMT correlations}} \longrightarrow \boxed{\text{high-fre}}$$

The middle Fourier arrow is exact by Theorems 16.1–16.2; the load-bearing open problem is the trace estimate itself. In particular, fixed powers, finitely many moments, or even a completed fourth-moment theorem cannot be silently promoted to microscopic BGS.

18 Reproducibility and adversarial checks

The maintained repository uses independent failure modes:

- Python/SymPy regenerates exact resultants, discriminants, real-trace subresultant ideals, finite-extension Frobenius checks, Jacobi/Cayley trace identities and elliptic spike algebra;
- C++20 independently checks finite-field hot paths and the 28-digit arbitrary-precision witness;
- Lean 4 kernel-checks finite pair-key/Cayley statements, historical falsifiers and the bounded Fibonacci-ratio residue;
- CI rebuilds all maintained code and recompiles this manuscript; a separate synchronization workflow rebuilds the committed PDF from the TeX source.

The dehomogenized row-support route supplies an important adversarial lesson: its first coefficient ideal was a safe one-way obstruction but not an exact geometric support because leading degree could drop at the projective point at infinity. The current Λ-ideal fixes the representation before applying subresultants. Likewise, finite Frobenius and ratio-orbit computations are regressions/falsifiers, not substitutes for the universal proofs.

19 Conclusion

The fourth-moment programme now has a sharper obstruction architecture. The pair-key reduction is exact; the mixed and corrected off/off terms are governed by the same Cayley additive energy; prime-order characteristic-zero collisions are rigid; canonical resultants and discriminants organize exceptional reductions; and the fixed-product-row problem has an exact order-three split support in the maximal real cyclotomic field. Outside that support the desired quadratic collision scale is already proved. The remaining pointwise split problem is a growing-K divisor theorem for the exact support, not another generic incidence estimate.

The minimal-extension theorem reveals a complementary arithmetic mechanism: nontrivial residue degree forces whole Frobenius packets of collisions. Combined with the same $O(K^4)$ resultant budget, this gives a density-one exact baseline through $\theta < 1/3$ for unrestricted prime pairs, while correctly leaving the split slice unchanged.

On the spectral side, the local-statistics gap is no longer described merely as "some missing local law." Distinct-index correlations have an exact partition-Möbius trace formula, and Poisson summation shows that microscopic statistics test trace powers at the Heisenberg scale. The next spectral target is therefore a concrete weighted joint-trace comparison for the correct symmetry-resolved quantum map, together with a high-frequency tail theorem. Microscopic BGS/GOE/COE universality remains open.

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